

Enterprise Data Governance & Data Quality Audit Report

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1. Executive Summary

This document presents an end-to-end Data Governance and Data Quality audit conducted on the Enterprise Sales Dataset. The project involved transforming unmodeled flat-file data into a structured **Star Schema Data Model**, defining clear metadata classifications, establishing a standardized **Business Glossary**, and deploying an automated **Data Quality Dashboard** in Power BI.

Through this implementation, the Overall Data Health Index reached **86.5%**, successfully identifying critical data ingestion errors across departments for targeted remediation.

2. Business Glossary

The Business Glossary provides standardized, enterprise-wide definitions for key business entities and metric calculations to resolve ambiguity across business units.

Term	Business Definition	Data Owner	Technical Formula / Rule
Gross Revenue	The total gross value of a sales transaction prior to any discounts or deductions.	Sales & Finance	$\$Gross \geq 0\$$
Net Revenue	The net monetary amount realized from a sale after applying allowable promotional discounts.	Finance	$\$Net = Gross - Discount\$$

Term	Business Definition	Data Owner	Technical Formula / Rule
Active Customer	A registered customer entity that has performed at least one valid registration or transaction.	Marketing & CRM	Valid cust_id required

3. Technical Metadata & PII Classification

To ensure compliance with data protection standards, the data model has been decentralized into a Star Schema, accompanied by technical metadata and **Personally Identifiable Information (PII)** security classifications.

Table 1: Dim_Customer (Dimension Table)

- **Description:** Stores master customer profiles and identity data.

Column Name	Data Type	Key Type	Security / PII Level	Description
cust_id	String	Primary Key (1)	Public	Unique customer identification code.
full_name	String	Attribute	Confidential (PII)	Full legal name of the customer.
email	String	Attribute	Confidential (PII)	Primary electronic contact address.
birth_date	Date	Attribute	Highly Confidential (PII)	Date of birth for demographic profiling.

Table 2: Dim_Department (Dimension Table)

- **Description:** Contains organizational units responsible for data asset ownership.

Column Name	Data Type	Key Type	Security / PII Level	Description
dept_id	Integer	Primary Key (1)	Public	Surrogate primary key for departments.
dept_owner	String	Natural Key	Public	Organizational department name.

Table 3: Fact_Sales (Fact Table)

- **Description:** Records transactional revenue metrics and associated quality flags.

Column Name	Data Type	Key Type	Security / PII Level	Description
cust_id	String	Foreign Key (\$\star\$)	Internal	Relational key linking to Dim_Customer.
dept_owner	String	Foreign Key (\$\star\$)	Internal	Relational key linking to Dim_Department.
gross_revenue	Currency	Measure	Restricted	Raw transactional gross amount.

Column Name	Data Type	Key Type	Security / PII Level	Description
discount_amt	Currency	Measure	Restricted	Calculated or applied discount amount.
net_revenue	Currency	Measure	Restricted	Transaction net realization value.
Is_Complete_Name	Integer (0/1)	Validation Flag	Internal	Data quality indicator for name presence.
Is_Valid_Email	Integer (0/1)	Validation Flag	Internal	Quality flag for standardized email syntax.
Is_Valid_Revenue	Integer (0/1)	Validation Flag	Internal	Quality flag ensuring non-negative gross value.
Is_Valid_net_Logic	Integer (0/1)	Validation Flag	Internal	Quality flag verifying business rule logic.

4. Data Quality Audit Log & Remediation Strategy

An exhaustive audit of \$1,500\$ records revealed key operational issues affecting data integrity:

Identified Issues Log

1. Whitespace & Empty String Anomaly (Completeness):

- **Issue:** Blank entries encoded as empty text ("") rather than formal SQL nulls, causing false completeness evaluation.
- **Remediation:** Applied Power Query Trim transformations and replaced empty strings with proper null values.

2. Syntax Corruption in Email Formatting (Validity):

- **Issue:** Email strings containing structural syntax errors (e.g., _at_ instead of @).
- **Remediation:** Flags raised via conditional column parsing; bad records routed for automated CRM data cleaning.

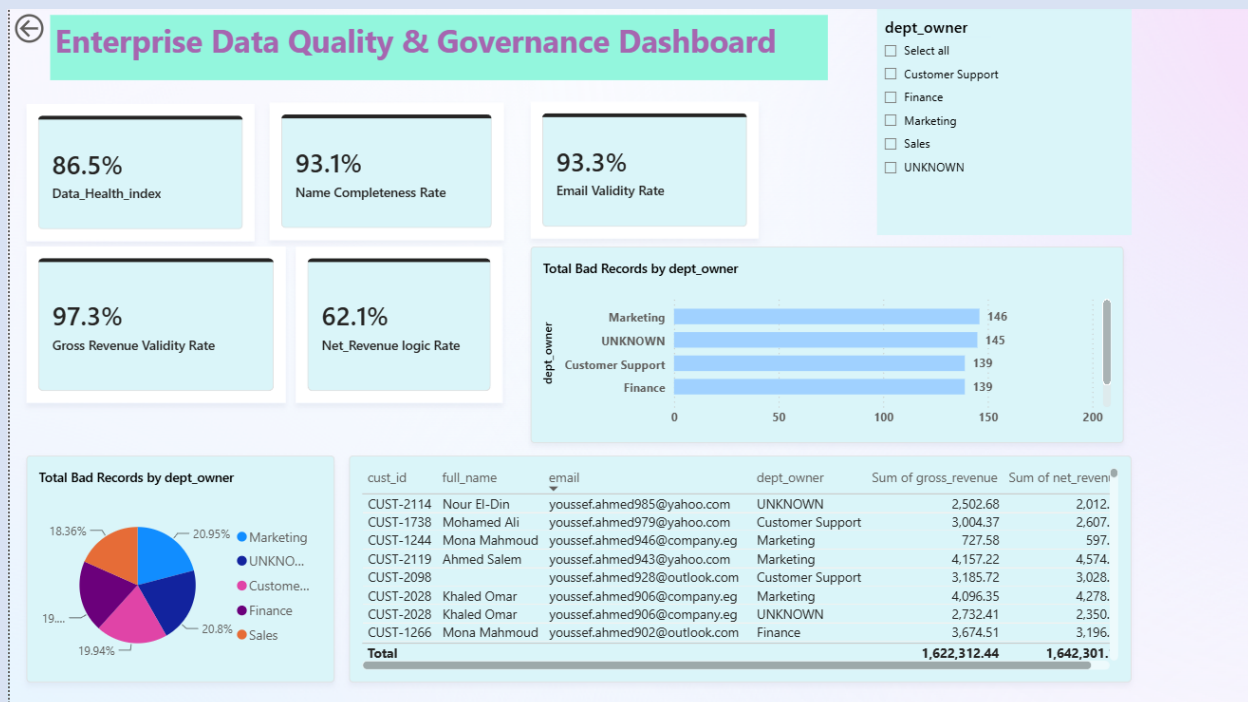
3. Cross-Field Data Mismatch (Consistency):

- **Issue:** Inconsistencies between customer names and associated email handles (e.g., mismatched identity records).
- **Remediation:** Assigned to Master Data Management (MDM) team for identity resolution.

4. Business Rule & Sign Inversion Errors (Accuracy):

- **Issue:** System logic errors causing negative gross values and inaccurate net revenue calculations.
- **Remediation:** Implemented DAX audit measures and flagged bad transactional rows.

5. Interactive Data Quality Dashboard (Power BI)



Executive Summary of Key Metrics:

- Overall Data Health Index: **86.5%**
- Name Completeness Rate: **93.1%**
- Email Validity Rate: **93.3%**
- Gross Revenue Validity Rate: **97.3%**
- Net Revenue Logic Rate: **62.1%**

Operational Insight: The primary driver of lower data quality scores stems from **Net Revenue Logic Errors (62.1%)**, predominantly concentrated within the **Marketing** and **UNKNOWN** departmental units. Remediation tasks have been logged in the Actionable Table visual for immediate Steward intervention.